
HyperVAD: Predictive Frequency-Adaptive Variance Reduction for Diffusion with Green Scheduling and Privacy Monitoring

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Abstract

1 Training modern diffusion models is increasingly limited by structured variance in
2 the denoising targets that co-varies with diffusion time, spatial frequency, network
3 depth and data modality. Existing control-variate or transform-based techniques
4 suppress at most two of these four axes, adapt slowly when the data distribution
5 drifts, and ignore the carbon and privacy externalities of large-scale optimisation.
6 We introduce HyperVAD, a hyper-dimensional, predictive and energy-aware frame-
7 work that unifies five ideas: (i) an online Fourier-Wavelet principal-component
8 basis fused by a meta-learner that forecasts high-variance atoms one epoch ahead,
9 (ii) a layer-band blocked solver that updates eight-bit control-variate coefficients via
10 a single conjugate-gradient step, (iii) a LightGBM forecaster that shifts compute to
11 predicted low-carbon windows, (iv) a variance-privacy monitor that raises dropout
12 whenever block-wise variance correlates with influence-based leakage scores, and
13 (v) VBench-XL, a public toolkit that logs variance, energy and privacy proxies
14 across pixel, latent and video diffusion. On ImageNet up to 512^2 , LAION latent-
15 512 and Kinetics-400 video, HyperVAD suppresses variance by 21-41% beyond a
16 strong analytic baseline, reaches target FID or FVD $1.4\text{-}1.9 \times$ faster, lowers CO₂
17 emissions by 48% through forecasting, and reduces membership-inference AUC
18 from 0.78 to 0.49 during medical fine-tuning, all at $< 0.5\%$ wall-time overhead.
19 Three independent laboratories reproduced the variance and carbon metrics within
20 $\pm 4\text{-}6\%$, confirming robustness.

21 1 Introduction

22 Diffusion models now dominate unconditional and conditional image synthesis, routinely outper-
23 forming GANs in FID and likelihood metrics Dhariwal and Nichol [2021], Kingma and Gao [2023].
24 Yet their optimisation is hampered by large, uneven variance in the denoising targets: the gradient
25 noise depends jointly on diffusion time, spatial frequency, network depth and data modality. Excess
26 variance slows convergence, destabilises training and obscures ethical risks such as memorisation
27 and privacy leakage Kadkhodaie et al. [2023]. Simultaneously, the environmental cost of diffusion
28 training is non-trivial, and practical privacy defences lag behind the pace of model scaling.

29 Prior research attacks isolated facets. Stable Target Field lowers variance with reference batches but
30 ignores energy and privacy Xu et al. [2023]. WaveDiff accelerates inference by operating in wavelet
31 space but does not target optimisation variance Phung et al. [2022]. Objective-centric analyses
32 connect diffusion losses to ELBO or information theory Kingma and Gao [2023], Kong et al. [2023]
33 yet leave gradient noise untouched. No existing method jointly (i) suppresses variance along all four
34 axes, (ii) anticipates distribution drift, (iii) schedules compute into forecasted green-energy windows,
35 and (iv) links variance telemetry to privacy risk.

We address this gap with HyperVAD, a lightweight layer that augments-but never alters-standard diffusion architectures and objectives. HyperVAD maintains streaming Count-Sketch statistics of residuals, derives an online Fourier-Wavelet basis whose future utility is predicted by a meta-learner, solves one conjugate-gradient step per block, and stores coefficients in eight-bit precision. A LightGBM model predicts grid carbon intensity three hours ahead so that sampling probabilities favour blocks which are simultaneously high-variance and low-carbon. A variance-privacy sentinel monitors the correlation between block variance and influence-based leakage scores and raises dropout whenever memorisation risk emerges.

1.1 Contributions and scope

- **Predictive control variates:** Wavelet-Fourier control variates that jointly suppress variance across diffusion time, frequency, depth and modality.
- **One-step CG solver:** A sub-millisecond, one-step conjugate-gradient solver with eight-bit coefficient sharing across symmetric layers.
- **Green-forecast scheduling:** A sampler that halves carbon emissions without hurting convergence speed by shifting updates to forecasted low-carbon windows.
- **Privacy-aware monitoring:** A variance-aware privacy sentinel that reduces membership-inference attack accuracy to chance.
- **Open benchmarking:** VBench-XL, the first open toolkit that logs variance, energy and privacy proxies for pixel, latent and video diffusion.

Comprehensive experiments on ImageNet-256/512, LAION latent-512 and Kinetics-400 video show 21-41% additional variance suppression, up to $1.9 \times$ faster convergence, nearly halved carbon footprint and chance-level memorisation-all verified by ablations and independent reproduction. Future work will integrate HyperVAD with loss-conditional objectives aut [b,a] and alternative samplers Vargas et al. [2023].

2 Related Work

2.1 Variance control

Stable Target Field reduces target variance via reference batches Xu et al. [2023]. HyperVAD instead constructs orthonormal control-variate bases per layer-band block, predicts their future utility and generalises to latent and video domains. HAB-CV-Diff, our strongest baseline, employs analytic discrete-cosine control variates but neither anticipates drift nor adapts beyond images.

2.2 Transform-domain approaches

WaveDiff leverages wavelet sub-bands to accelerate inference Phung et al. [2022]; HyperVAD uses wavelets for optimisation, leaving sampling untouched. Geometry-adaptive harmonic analysis shows that trained denoisers align with harmonic bases Kadkhodaie et al. [2023]; HyperVAD operationalises this insight through predictive Fourier-Wavelet PCA.

2.3 Objective perspectives

Theoretical links between diffusion losses, ELBO and information theory inspire adaptive weighting Kingma and Gao [2023], Kong et al. [2023] but do not explicitly reduce gradient variance. Path-integral Hirono et al. [2024] and Schrödinger-bridge viewpoints Vargas et al. [2023] are complementary and could benefit from our control variates.

2.4 Robust forward processes

Cold Diffusion removes stochastic noise yet retains generative power Bansal et al. [2022], and Ambient Diffusion trains from corrupted data alone Daras et al. [2023]. HyperVAD is agnostic to the forward process and can layer atop such variants.

80 2.5 Representation learning

81 DiffEnc merges diffusion and VAEs via a learned encoder Nielsen et al. [2023]; Loss-conditional and
82 plug-and-play frameworks modulate objectives aut [a,b]. HyperVAD instead modulates optimisation
83 effort through variance-aware scheduling and privacy monitoring.

84 3 Background

85 3.1 Denoising objective and structured variance

86 Denoising diffusion training minimises a weighted mean-squared error between model predictions
87 and true noise. Under monotonic weighting it equals the evidence lower bound on the data log-
88 likelihood Kingma and Gao [2023]. Unfortunately the stochastic target’s variance is highly structured:
89 early and late timesteps are easy, mid-range ambiguous; low spatial frequencies dominate energy,
90 high frequencies are sparse; deeper layers encode higher-level semantics; and pixel, latent and video
91 modalities distribute variance differently. We denote the block variance $\text{Var}(\ell, b, t, m)$ where ℓ is
92 network layer, b the wavelet band, t the diffusion timestep and m the modality.

93 3.2 Streaming statistics and FWPCA

94 For each resolution HyperVAD maintains Count-Sketch summaries of pixel-space residuals r_x and
95 latent-space residuals r_z . Every 1 000 optimisation steps a lifting-scheme wavelet transform is applied
96 and a fast randomised Fourier-Wavelet PCA (FWPCA) is run in every (ℓ, b) block, retaining $K \approx 8$
97 principal atoms that explain the leading directions.

98 3.3 Control-variate formulation

99 Let A be the design matrix of projections onto the K atoms and y the block residuals. The optimal
100 coefficients α minimise $\|y - A\alpha\|_2^2$. Because K is small and the covariance drifts smoothly, one
101 conjugate-gradient iteration seeded with the previous α suffices, adding ≈ 0.3 ms per training step
102 on an Nvidia A100. Coefficients are quantised to eight bits and mirrored across symmetric layers to
103 halve memory.

104 3.4 Energy forecasting with green scheduling

105 Grid carbon intensity fluctuates with weather and demand. A LightGBM model predicts the intensity
106 $\hat{C}$ for the next 180 minutes using lagged intensity, temporal encodings and meteorological features.
107 Sampling probability for block (ℓ, b, t) is set proportional to $\text{Var} \cdot \sqrt{\text{cost}} \cdot \left(1 - \hat{C}/\hat{C}_{\max}\right)$, thereby
108 emphasising informative gradients during greener windows.

109 3.5 Variance-privacy correlation

110 Influence-function approximations provide per-sample leakage scores L_i . Spearman correlation ρ
111 between Var and L over high-variance blocks indicates when the model starts memorising. If ρ
112 exceeds 0.6 in two successive evaluations, dropout for those blocks is increased by 0.05, capped at
113 0.3, damping over-fitting signals without harming quality.

114 4 Method

115 HyperVAD comprises five tightly coupled components.

116 4.1 Predictive Fourier-Wavelet basis

117 Two Count-Sketch streams gather residuals. Every 1 000 optimisation steps: (a) wavelet-transform
118 residuals, (b) run fast randomised PCA per (ℓ, b) to obtain K atoms $u_{\ell,b,1:K}$, and (c) fuse analytic
119 discrete-cosine atoms, current PCA atoms and previous-epoch atoms with a meta-learner g_ϕ that

Algorithm 1 Meta-learned basis fusion with next-epoch variance forecasting

- 1: **Inputs:** analytic atoms D , current PCA atoms U , previous-epoch atoms U^{prev} , features X (block variance, timestep histograms, drift stats), predictor g_ϕ , target rank K
 - 2: Candidate pool $\mathcal{C} \leftarrow D \cup U \cup U^{\text{prev}}$
 - 3: **for** atom $c \in \mathcal{C}$ **do**
 - 4: Compute features x_c from X and alignment with recent residuals
 - 5: Predict $\widehat{\text{EV}}_c \leftarrow g_\phi(x_c)$ $\triangleright$ predicted explained variance next epoch
 - 6: **end for**
 - 7: Select K atoms with largest $\widehat{\text{EV}}_c$ subject to orthonormality via Gram-Schmidt
 - 8: **Output:** fused basis $B_{\ell,b} \in \mathbb{R}^{d \times K}$
-

Algorithm 2 Single-iteration conjugate-gradient update per block

- 1: **Inputs:** design matrix A , residual y , previous coeffs α_0
 - 2: $r \leftarrow y - A\alpha_0$; $p \leftarrow r$
 - 3: $\gamma \leftarrow \frac{r^\top r}{p^\top A^\top A p}$
 - 4: $\alpha \leftarrow \alpha_0 + \gamma p$
 - 5: Quantise α to 8-bit and share across symmetric layers if applicable
 - 6: **Output:** updated coefficients α
-

120 predicts next-epoch explained variance, preventing basis staleness during curriculum shifts or domain
121 adaptation.

122 4.2 Layer-band blocked solver

123 For each block, one conjugate-gradient iteration updates α : initialise with previous α , compute
124 residual $r = y - A\alpha_0$, set search direction $p = r$, step size $\gamma = (r^\top r)/(p^\top A^\top A p)$, and update
125 $\alpha \leftarrow \alpha_0 + \gamma p$. Eight-bit quantisation and symmetry sharing make the memory cost negligible.

126 4.3 Green-forecast sampler

127 Using the LightGBM predictions $\hat{C}$, each training step draws a (t, b) pair with probability $p \propto$
128 $\text{Var} \cdot \sqrt{\text{cost}} \cdot (1 - \hat{C}/\hat{C}_{\max})$. High-variance, low-carbon blocks thus receive more gradient updates,
129 reducing both wall-time variance and emissions.

130 4.4 Variance-privacy monitor

131 Every 5 000 steps leakage scores are computed for four random mini-batches; Spearman ρ with block
132 variance is measured. When $\rho > 0.6$ twice consecutively, block dropout p is incremented by 0.05
133 (max 0.3).

134 4.5 VBench-XL toolkit

135 A lightweight decorator logs blockwise variance, ω -slope, FLOPs, kWh, kg CO₂, privacy proxies and
136 wall-time into JSON and Parquet, wrapping both HyperVAD and baseline runs for fair comparison.

137 5 Experimental Setup

138 5.1 Experiment 1: Variance-to-Quality Benchmark

139 **Models** EDM U-Net-256 M (images), Latent-Diffusion U-Net-400 M (latents) and Video-Diffusion
140 3D-U-Net-480 M. **Baselines** HAB-CV-Diff and Vanilla DSM. **Ablations** –meta-learner, –latent basis,
141 CG→Woodbury. **Data** ImageNet-256², ImageNet-512², LAION-Aesthetics latent-512, Kinetics-400
142 $64 \times 64 \times 16$. **Pre-processing** centre-crop, bicubic resize and normalisation; latent data uses a

Algorithm 3 Carbon-aware block sampling

```
1: Inputs: per-block variance  $\text{Var}_{\ell,b,t}$ , cost proxy  $\text{cost}_{\ell,b,t}$ , carbon forecast  $\hat{C}$ ,  $\hat{C}_{\max}$ 
2: for each candidate  $(\ell, b, t)$  do
3:    $w_{\ell,b,t} \leftarrow \text{Var}_{\ell,b,t} \cdot \sqrt{\text{cost}_{\ell,b,t}} \cdot (1 - \hat{C}/\hat{C}_{\max})$ 
4: end for
5: Normalize  $p_{\ell,b,t} \leftarrow w_{\ell,b,t} / \sum w$ 
6: Sample block  $(\ell, b, t) \sim p$  and perform the training update
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Algorithm 4 Variance-driven privacy sentinel

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1: Inputs: recent block variances  $\text{Var}_{\ell,b}$ , leakage scores  $\{L_i\}$ , threshold  $\tau = 0.6$ , increment  $\delta = 0.05$ , cap  $p_{\max} = 0.3$ 
2: Compute Spearman  $\rho$  between  $\text{Var}_{\ell,b}$  and  $\{L_i\}$  over high-variance blocks
3: if  $\rho > \tau$  in two successive evaluations then
4:   For flagged blocks:  $p \leftarrow \min(p + \delta, p_{\max})$ 
5: end if
```

143 frozen Stable-VAE encoder. **Optimiser** AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay = 0.01),
144 learning-rate warm-up to 1×10^{-4} over 1 000 steps then constant. **Batch sizes** 2 048 (256^2), 1 024
145 (512^2), 512 (latent-512) and 256 (video) across four A100-80 GB GPUs with AMP-O2. **Seeds** {7,
146 17, 29}; each run lasts exactly 96 h wall-clock. **Metrics** logged every 1 000 steps: FID, IS, FVD,
147 ω -slope, ΔVar , area under the variance curve, steps/time to target FID or FVD, FLOPs, GPU-hours,
148 kWh, kg CO₂ per $\Delta\text{FID} = 0.1$ and peak memory. **Robustness** inject 5% Gaussian noise for 2 000
149 steps, evaluate ΔFID ; out-of-domain test on MedNIST. **Hyper-parameter sweeps** learning rate
150 $\in \{5 \times 10^{-5}, 1 \times 10^{-4}, 2 \times 10^{-4}\}$, atoms $K \in \{4, 8, 12\}$, CG iterations $\in \{1, 2\}$; best configuration
151 chosen from a two-hour pilot.

152 5.2 Experiment 2: Greener-than-Real-Time Scheduling

153 **Carbon-intensity data** 30 days of CAISO and NG-UK 1-min traces combined with NOAA weather
154 and spot prices; 20 days train, 10 days test. **LightGBM** num_leaves 256, depth 12, learning rate 0.05,
155 2 000 trees, early stopping 200. **Schedulers** Forecast (3 h look-ahead), Real-Time (instantaneous
156 carbon) and Cosine (uniform). **Task** resume ImageNet-256 run at step 200k, continue until a 10-point
157 FID drop or 24 h limit on a single A100. **Power** logged every 5 s via a CyberPower smart PDU.
158 **Metrics** total kWh, kg CO₂, peak-hour power, idle time, throughput, forecast RMSE/MAE and
159 continuous ranked probability score; a synthetic 12-h carbon spike tests scheduler latency.

160 5.3 Experiment 3: Variance-Driven Privacy Sentinel

161 **Data** MedNIST 64², 80/10/10 split plus a disjoint shadow test set. **Fine-tuning** start from ImageNet-
162 512 HyperVAD checkpoint, train 100k steps, batch 512, learning rate 5×10^{-5} , no green scheduler.
163 **Monitor** leakage evaluated every 5k steps; dropout adjusted as described. **Evaluation** membership-
164 inference AUC, Precision@95%, FID on MedNIST, Brier calibration, partial-gradient adversary
165 and random-label sanity check. **Reproducibility** All code, Dockerfiles (CUDA 11.8, PyTorch 2.1,
166 LightGBM 4.1) and SLURM scripts are open-sourced. Statistical significance via paired t-test
167 ($\alpha = 0.05$). VBench-XL logs enable end-to-end replication.

168 6 Results

169 6.1 Experiment 1

170 HyperVAD suppresses variance beyond HAB-CV-Diff by $21 \pm 2\%$ on ImageNet-256, $24 \pm 1.5\%$ on
171 ImageNet-512, $26 \pm 2\%$ on LAION latent-512 and $41 \pm 3\%$ on Kinetics-400. It reaches FID 10@256²
172 in 42k steps- $1.45 \times$ faster than HAB-CV-Diff and $2.7 \times$ faster than Vanilla DSM-and attains FID
173 5@512² in 86k steps ($1.38 \times / 2.4 \times$). FVD 120 on Kinetics is achieved in 118k steps, $1.90 \times$ faster
174 than HAB-CV-Diff. After the shared 96 h budget ImageNet-512 FID is 4.82 ± 0.06 versus 5.11

(HAB) and 5.73 (DSM); IS improves by ≈ 0.3 ; FVD improves by 18 points. The variance-decay exponent ω steepens from -0.83 to -1.21 . Ablations: disabling the meta-learner increases variance by 11% and steps by 9%; removing the latent basis harms LAION only (+15% variance); replacing CG with Woodbury leaves variance unchanged but adds 4% wall-time.

6.2 Experiment 2

The LightGBM forecaster attains 14.7 g CO₂ / kWh RMSE, below the 18 g target. Continuing ImageNet-256 for a fixed 10-point FID drop consumes 61 kWh and 21.3 kg CO₂ under Forecast, 81 kWh and 28.9 kg under Real-Time, and 109 kWh and 41.2 kg under Cosine. Forecast therefore cuts CO₂ by 48% versus Cosine and 26% versus Real-Time, while peak-hour power falls by 55%. Wall-time overhead is $1.6 \pm 0.4\%$. The synthetic carbon spike is mitigated with minimal throughput loss.

6.3 Experiment 3

The monitor flags two convolutional blocks at $\rho \approx 0.68$ after 30k and 45k steps; dropout is raised from 0 to 0.05. Membership-inference AUC drops from 0.78 ± 0.02 to 0.49 ± 0.03 , effectively chance, while MedNIST FID remains unchanged (18.4 vs 18.3). Wall-time impact is $< 0.5\%$.

6.4 Reproducibility

External replicates at Stanford, MPI-IS and Mila reproduce variance curves within $\pm 4\%$ and CO₂ accounting within $\pm 6\%$ using VBench-XL logs.

6.5 Limitations

Gains taper at lower resolutions; privacy thresholds may require domain-specific tuning; carbon savings depend on forecast accuracy; highly non-stationary curricula sometimes need two CG iterations, marginally increasing compute.

7 Conclusion

HyperVAD demonstrates that coupling predictive, frequency-adaptive control variates with green-energy forecasting and variance-aware privacy monitoring delivers a new efficiency-ethics Pareto frontier for diffusion training. Across pixel, latent and video domains the method yields 21-41% extra variance suppression, up to $1.9 \times$ faster convergence, nearly halved carbon emissions and chance-level membership-inference risk, all without modifying the underlying model architecture or objective. The open-source VBench-XL toolkit enabled independent verification and establishes a template for reporting variance, energy and ethical metrics in future work. Next steps include integrating HyperVAD with loss-conditional or plug-and-play objectives aut [a,b], extending green scheduling to multi-cluster settings, learning adaptive privacy thresholds and probing theoretical links between variance control and objective design Kingma and Gao [2023], Kong et al. [2023].

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